

Kush Patel

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EDUCATION

Rutgers University, School of Arts & Science

Bachelor of Science in Computer Science – GPA: 3.62 / 4.0

New Brunswick, NJ

Jun '22 – May '24

Minor: Business Administrations

- **Relevant Coursework**

Artificial Intelligence, Deep Learning, Operating Systems, Computer Security, Systems Programming, Statistics, Design & Analysis of Computer Algorithms, Computer Networks (audit)

- **Honors & Achievement**

Cum Laude, Dean's List 2023, Final year 4.0 GPA

Middlesex County College

Associate of Science in Computer Science GPA: 3.78 / 4.0

Edison, NJ

Sept '19 – May '22

WORK EXPERIENCE

R&D Altanova, Advantest Group's

Technology Specialist, developing

South Plainfield, NJ

Jun '24 - Present

- Analyzed complex data sets of GL Impacts Transactions using PostgreSQL and pandas creating using data visualization
- Engineered a robust automation Production Planning & Scheduling Systems (PPSS) utilizing real time data, achieving 96% reduction from manual excel report, featuring multiuser access control, notifications, and dynamic scheduling
- Automated script updates by pulling capacitor and resistor values from Excel, eliminating manual editing of over 50,000 lines files implementing robust I/O operations, and improving workflow efficiency by 100%
- Developed a REST API & IP/TCP communication framework that connects existing software with a robotic arm, automating circuit measurements resulting increase in accuracy & efficiency.

R&D Altanova, Advantest Group's

Information Technology Intern

South Plainfield, NJ

Aug '22 - May '24

- Installed and configured over 50 desktops and laptops, along with peripheral devices, providing technical support to end-users, which resulted in a 15% decrease in hardware-related support tickets
- Managed Microsoft Azure administration, overseeing email configurations, Intune deployments, and user authentication setups to enhance system security and user experience
- Administered server domain permissions and share access, ensuring secure and efficient resource access for over 200 users, which enhanced operational productivity
- Developed and implemented an IT Asset Management system that streamlined inventory tracking and management of over 1,000 IT assets, resulting in reduced asset retrieval time by 60% and improved compliance audits
- Created a custom dashboard that integrated real-time analytics to track current and upcoming jobs, prioritized by arrival time, resulted in improving scheduling efficiency and reducing paper waste by 20%

BAPS Charities & Youth Organization - Volunteer

Edison Development Team

Edison, NJ

Nov '20 - Present

- Collaborated with a team to install cameras, fiber optic cables, and Ethernet networks, significantly improving event infrastructure and connectivity.
- Mentored 30 youths, organizing educational activities to teach over 100 children about Hindu arts and culture.
- Coordinated and managed food service logistics for events, serving over 900 attendees efficiently.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript

API Development: REST API, API

ML & AI: PyTorch, TensorFlow, sk-learn

Testing & Automation: Jest, Cucumber

Web Dev: HTML, CSS, React, Django, Flask, .NET

Database: MySQL, PostgreSQL, MongoDB

DevOps & Version Control: GitHub, CI/CD

Operating System: Linux, Windows, MacOS

PROJECTS

Scam Email Detection Model – Python, NPL, Pytorch, TensorFlow

Jun' 24 - Present

Self-Learning Project

- Trained a Scam Email Detection Model on a Kaggle dataset of over 100,000 emails, classifying content as 'scam' or 'real'
- Utilized PyTorch and advanced NLP techniques to optimize model performance and reduce false positives
- Implemented feature extraction methods like TF-IDF and word embeddings for improved classification accuracy of 63%
- Conducted EDA to identify patterns and anomalies, using statistical methods like correlation analysis and outlier detection to inform feature selection

Wes AI Chatbot – Python, transformers, Django, html, CSS, JS

Mar' 24

Rutgers University Hackathon HACKRU Spring 2024

- An interactive AI chatbot for Rutgers University, using Python and Django, integrating a sentence transformer model to provide real-time responses to campus-related inquiries
- Designed and implemented a user-friendly interface with HTML, CSS, and JavaScript, enabling students to easily access information on bus routes, course registrations, and campus services
- Collaborated in a team to manage project development using GitHub, enhancing code efficiency with GitHub and ensuring seamless integration of frontend and backend components

File System (FUSE) – C, Makefile, Benchmark, Report Analysis

Nov '23 – Dec'23

CSC-416 Operating Systems

- Developed a custom file system in C using FUSE, writing over 1,700 lines of code to implement core file operations like file creation, reading, and directory management
- Optimized inode allocation and block indexing, improving I/O efficiency for large data sets
- Implemented indirect pointers and directory removal, enhancing file system functionality, and passing all benchmark tests

Tripmate Notifier – Java, JavaFX, API

Oct '23

Rutgers University Hackathon HACKRU Fall 2023 – 1st NJ Transit Winner

- Developed Tripmate Notifier using Java and JavaFX, providing real-time train status alerts via lock-screen notifications to assist NJ Transit commuters with trip updates.
- Leveraged NJ Transit datasets to analyze and extract meaningful trip information, utilizing data cleaning techniques to enhance the accuracy and relevance of trip notifications.
- Designed and implemented the backend architecture to process live trip data and integrate notifications, ensuring seamless user interaction with real-time updates

EXTRA-CURRICULAR ACTIVITIES

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| • Google Cloud Get Certified Edition 3 | <i>Aug '24 – Present</i> |
| • New Jersey Softball Cricket League (NJSBCL) | <i>Feb '21 – Present</i> |
| • Rutgers University Mobile Application Development Club (RUMAD) | <i>Jan '23 – May '24</i> |
| • Attendee, Student Conference BAPS | <i>Oct '23</i> |
| • Math Calculus and Algebra 10 th Grade Tutor | <i>Nov '21 – May '22</i> |

ACHIEVEMENTS

- **150+ LeetCode Problems Solved** (Python, JavaScript, C)
- **Hackathon Winner:** Invited to present at NJ Transit headquarters.
- **Deployed 11+ Software Solutions** in 2023-24 at R&D Altanova.
- **Google Cloud Engineer Associate** (Pursuing)
- **Cisco Certified Ethical Hacker**
- **Cisco Cybersecurity Certification**